# Where you finish is not the whole trip

GUIDED NOTES / use with the projected diagram Name _____

## A SAY THE WORD

### Displacement / ES: desplazamiento / PT: deslocamento

What changed from start to finish? The sign gives the direction.

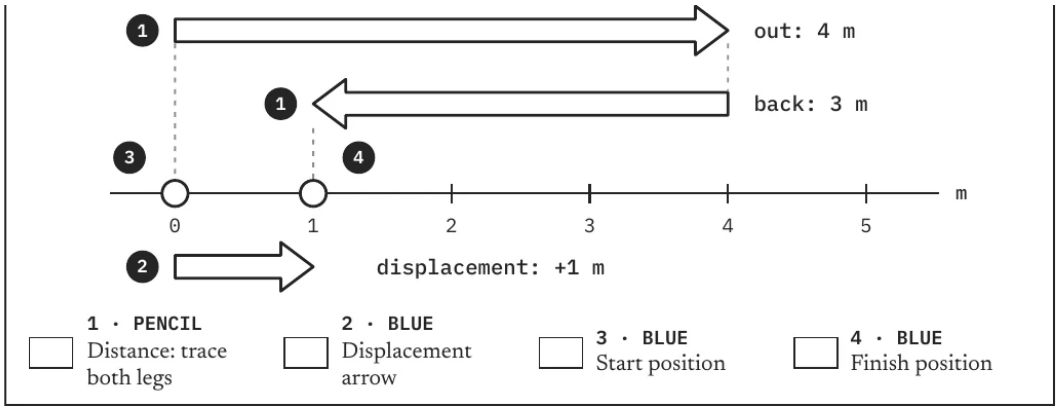
Also on your note page: Position, Distance, Scalar / vector.

## B READ THE RELATIONSHIP

### Displacement tells how far and which way the finish is from the start.

Displacement = final position − initial position

## C TRACE THE ORIGINAL FIGURE




Use the numbered key. Trace one feature. Tell your partner what it means.

## D FOLLOW THE WORKED EXAMPLE

**A cart goes from 2 m to 6 m, then back to 4 m. Right is positive. Find the distance and the displacement.**

**Given** x i = 2 m turn at 6 m x f = 4 m

**Rule** distance = leg 1 + leg 2 Δx = x f − x i

**Work** 4 m + 2 m 4 m − 2 m

**Answer** distance 6 m displacement +2 m

## E SAVE FOR THE FINAL MEETING

Before we can say where XJ is, we need a zero and a + direction. Analysts measure its position from Earth and track its displacement, not the winding path it travels.

# Read the physics: Displacement

SUPPORTED READING / read one part, then point to the evidence Name _____

## 01 PICTURE IT

Draw an origin: this is zero. Choose which way counts as positive. Now a position tells both people the same location.

## 02 HOW IT WORKS

Trace every part of the trip to find distance. For displacement, go straight from start to finish. Subtract the starting position from the final position. Keep the sign.

## 03 WHAT TO CHECK

The object can travel a long distance and end close to its start. If it returns to its start, displacement is zero.

## 04 TRY THE EXAMPLE FROM YOUR NOTES

A cart goes from 2 m to 6 m, then back to 4 m. Right is positive. Find the distance and the displacement. distance 6 m displacement +2 m.

## 05 CONNECT TO XJ

Before we can say where XJ is, we need a zero and a + direction. Analysts measure its position from Earth and track its displacement, not the winding path it travels.

SAY IT BEFORE YOU WRITE

Which two positions belong in the subtraction? The path is ___. The change from start to finish is ___.

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Position

ES: posición / PT: posição

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain position to a friend?<br>In my words, this means ___. | The doorway is zero and east is positive. Where is a cart at −3 m?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| How would you mark a cart 2 m east of the doorway on a number line?<br>My example shows ___. | "I walked 7 m." Do we know your position now?<br>I would change ___ because ___. |

## Distance

ES: distancia / PT: distância

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain distance to a friend?<br>In my words, this means ___. | A cart goes 3 m right and 2 m left. How much ground do its wheels cover?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Draw a trip that covers 6 m and finishes at its starting point.<br>My example shows ___. | A partner writes −5 m for distance. What does the minus sign tell you?<br>I would change ___ because ___. |

# Frayer models

SECTION A / VOCABULARY Name _____

Say your idea in any language. Write, draw, or label it in the box.

## Displacement

ES: desplazamiento / PT: deslocamento

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain displacement to a friend?<br>In my words, this means ___. | Which two marks tell you the displacement even if the cart turns around?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Go from 2 m to 6 m and back to 4 m. What single arrow shows the displacement?<br>My example shows ___. | Someone adds both legs and gets 6 m. What did they find?<br>I would change ___ because ___. |

## Scalar / vector

ES: escalar / vector / PT: escalar / vetor

| Meaning in my words | How can you tell? |
|---|---|
| How would you explain the difference between a scalar and a vector?<br>In my words, this means ___. | Which tells you which way to draw an arrow: 5 m or 5 m west?<br>I can tell because ___. |
| **Example or sketch** | **Non-example and why** |
| Give a scalar and a vector for a 4 m walk straight east.<br>My example shows ___. | Two students each walk 4 m, one east and one west. Are their displacements equal?<br>I would change ___ because ___. |

# Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

## Put start, turn, and finish on a number line. Label distance on the path and displacement on the shortcut.

Labels you can use: start · finish · direction.

Choose zero, direction, and scale.



**Tell a partner**

The path is ___. The change from start to finish is ___.

# Send the rover home

02

**SECTION E** FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

**Our goal:** Tell distance traveled and change in position apart.

**CHOOSE ONE JOB TOGETHER**

**ROUTE ENGINEER** ☐

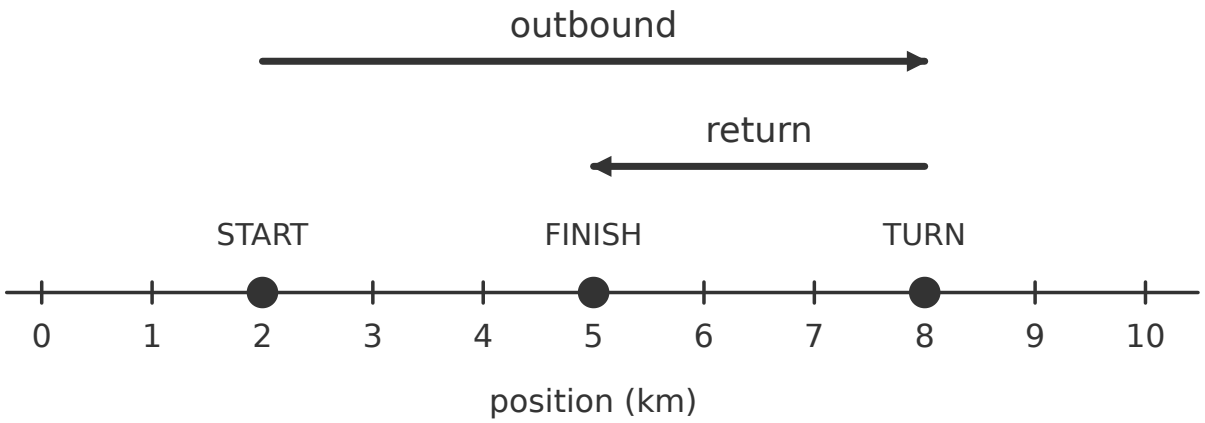
Mark both parts of the route and draw one start-to-finish arrow. Report the trip length and change in position.

**RADIO DISPATCHER** ☐

Guide a new driver to the finish. Give the trip length, change in position, and final location.

**Word help:** route / distance / displacement

A survey rover tracks XJ's model from **+2** to **+8** to **+5 km**. Dispatch needs both mileage and final location.




## 01 OUR FIRST CALL

**What numbers should dispatch use? Say what each number measures.**

The trip was ___. The finish is ___.

## 02 OUR REVISED CALL

**Wait for the reveal. Keep your first call.**

New clue: _____

Check the new location instruction. What should dispatch tell the driver now?

Now we would say ___ because ___.

**Switch speakers.** Does that number tell the driver how far to travel or where to finish?

| ☐ GREEN | ☐ RED | ☐ YELLOW | ☐ BLUE | ☐ PURPLE | ☐ ORANGE | ☐ PENCIL |
|---|---|---|---|---|---|---|
| speeding up | slowing down | steady / balanced | position / displacement | velocity | forces | other marks |